

OSM Geocoder Sheet Task Audit

Sheet 1, Sheet 2, and Sheet 3 task checks on the Europe binary snapshot

Scope

This report checks the final local OSM implementation task by task. Every task section contains ten distinct checks or queries. Runtime checks call the local HTTP server. Static checks use the server statistics and source-code behavior where a task is not naturally a single query.

Dataset And Server

Snapshot	/home/ganesh/europe-geocoder-v007.bin
Houses	104067059
Streets	16721783
POIs	13681952
Admin areas	433064
Forward index terms	3061667
Primary index terms	3049324
Server spatial index build	32692.074 ms
Embedded forward index rebuild time	0.0 ms
Dataset memory estimate	20874.15 MB

Result Summary

NOT IMPLEMENTED	30
PASS	199
WARN	1

Browser Recheck

The complete task audit was rerun through the local HTTP server before this PDF was compiled. A second browser-only replay was attempted against the saved probe URLs. The browser automation layer loaded / normally, but blocked raw /api/... page navigations with ERR_BLOCKED_BY_CLIENT. Because of that browser-side restriction, the live browser confirmation was done through the real frontend controls: search input, LLM checkbox, type-filter buttons, and Search button. These checks exercise the same user path that is used during a demo.

No.	Browser query	UI mode	Verdict	Observed frontend result
1	Stuttgart	Deterministic All	PASS	buttons=20; wallMs=725; first=area: Stuttgart
2	stuttgart	Deterministic All	PASS	buttons=20; wallMs=755; first=area: Stuttgart
3	stuttgrat	Deterministic All	PASS	buttons=20; wallMs=722; first=area: Stuttgart
4	berli	Deterministic All	PASS	buttons=20; wallMs=767; first=area: Berlin
5	koenigstr	Deterministic All	PASS	buttons=20; wallMs=723; first=street: Königstraße
6	Koenigstrasse Stuttgart	Deterministic Streets	PASS	buttons=20; wallMs=732; first=street: Königstraße
7	Stuttgart	Deterministic Admin	PASS	buttons=9; wallMs=738; first=area: Stuttgart
8	Burger King Stuttgart	Deterministic POIs	PASS	buttons=20; wallMs=732; first=poi: Burger King
9	Oxford Street London	Deterministic Streets	PASS	buttons=20; wallMs=716; first=street: Oxford Street
10	Aalen Bahnhofstrasse 10	Deterministic Houses	PASS	buttons=2; wallMs=723; first=house: Bahnhofstraße 10
11	Where can I buy shampoo near hotels in Stuttgart	LLM POIs	PASS	buttons=20; wallMs=2536; first=LLM: Find shampoo-selling personal care shops near hotels in Stuttgart.
12	Where can I find bread near hotels in London	LLM POIs	PASS	buttons=20; wallMs=2135; first=LLM: Find bread or bakery POIs close to hotels in London.
13	Where can I find park near hotels in Berlin	LLM POIs	PASS	buttons=20; wallMs=1820; first=LLM: Find parks near hotels in Berlin
14	where can i find burger king	LLM POIs	PASS	buttons=20; wallMs=1404; first=poi: Burger King
15	wo bekomme ich Hustensaft	LLM POIs	PASS	buttons=20; wallMs=1798; first=LLM: Find likely medicine shops near the given address.

Main Findings

- Fuzzy search now uses suffix-array seed lookup plus bounded edit-distance verification, so it does not depend on scanning the full Europe vocabulary.

- One-token place ranking gives stronger weight to important admin levels, exact city/place matches, prefix matches, and German umlaut-missing variants.
- The geocoder result list supports mixed ranking by default and type-filtered ranking for houses, streets, admin areas, and POIs.
- Street-like partial tokens such as koenigstr are handled as street candidates instead of being promoted as administrative place names.
- Some famous street queries can still be mixed with POI names in the all-results list; the type-filtered street view exposes the street ranking directly.
- Sheet 2 optional polygon overlay questions, Sheet 3 aggregation, and Sheet 3 strict results-only-in-view filtering are not implemented in the final program.

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Sheet 1

1.1 Mandatory Task 1: Understanding OSM Data Structures

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	nodes scanned counter present	PASS			housesFromNodes=49868010
2	ways converted to houses	PASS			housesFromWays=54060704
3	relations converted to houses	PASS			housesFromRelations=138345
4	street records stored	PASS			streets=16721783
5	admin areas stored	PASS			adminAreas=433064
6	POI records stored	PASS			pois=13681952
7	admin parent links present	PASS			adminParentAreaLinks=1769774
8	house-admin links present	PASS			houseAdminAreaLinks=580694902
9	street-admin links present	PASS			streetAdminAreaLinks=94117926
10	POI-admin links present	PASS			poiAdminAreaLinks=76671823

1.2 Mandatory Task 2: Extract Relevant Data from PBF

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	houses total greater than zero	PASS			houses=104067059
2	houses missing street counted	PASS			housesMissingStreet=14526295
3	houses missing number counted	PASS			housesMissingNumber=3365472
4	POIs from nodes counted	PASS			poisFromNodes=5165848
5	POIs from ways counted	PASS			poisFromWays=8516104
6	Stuttgart houses extraction /api/houses	PASS	27.010		houses=80; truncated=True
7	Berlin streets extraction /api/streets	PASS	25.310		streets=80; truncated=True
8	Paris admin extraction /api/admin	PASS	12.193		areas=80; truncated=True
9	London houses extraction /api/houses	PASS	17.245		houses=80; truncated=True
10	Vienna streets extraction /api/streets	PASS	7.657		streets=80; truncated=True

1.3 Mandatory Task 3: Timings and Stats

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	boundary relation scan timing	PASS			relationScan=22.83 s
2	dataset extraction timing	PASS			extraction=1436.281 s
3	boundary assembly timing	PASS			relationAssembly=4.074 s
4	admin lookup timing	PASS			adminLookup=2757.994 s
5	street connection timing	PASS			streetConnection=25.819 s
6	total parse timing	PASS			totalSeconds=4246.998
7	input throughput	PASS			inputMBps=7.69
8	dataset memory estimate	PASS			datasetMB=20874.15
9	RSS peak memory recorded	PASS			rssPeakMB=15997.8
10	forward index memory estimate	PASS			geocodeIndexEstimatedMB=7746.31

1.4 Mandatory Task 4: Visualization Frontend and Viewport Data

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	Stuttgart /api/houses	PASS	9.044		houses=80; truncated=True
2	Berlin /api/streets	PASS	4.655		streets=80; truncated=True
3	Muenchen /api/admin	PASS	32.524		areas=80; truncated=True
4	Hamburg /api/houses	PASS	24.787		houses=80; truncated=True
5	Paris /api/streets	PASS	15.459		streets=80; truncated=True

6	London /api/admin	PASS	14.453		areas=17; truncated=False
7	Wien /api/houses	PASS	19.975		houses=80; truncated=True
8	Zuerich /api/streets	PASS	27.509		streets=80; truncated=True
9	Barcelona /api/admin	PASS	14.329		areas=72; truncated=False
10	Amsterdam /api/houses	PASS	36.573		houses=80; truncated=True

1.5 Optional Task 1: Connect Streets

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	Koenigstrasse Stuttgart	PASS	74.447	56.077	count=5; first=Königstraße; mode=primary; typeFilter=all
2	Kurfuerstendamm Berlin	PASS	28.381	2.540	count=5; first=Kurfürstendamm; mode=primary; typeFilter=all
3	Champs Elysees Paris	PASS	54.487	56.715	count=5; first=Radisson Blu Hotel Champs Elysees; mode=primary; typeFilter=all
4	Oxford Street London	PASS	55.631	38.993	count=5; first=Oxford Street; mode=primary; typeFilter=all
5	Mariahilfer Strasse Wien	PASS	593.873	630.278	count=5; first=Mariahilfer Straße; mode=primary; typeFilter=all
6	Bahnhofstrasse Zuerich	PASS	371.344	399.865	count=5; first=Bahnhofstrasse; mode=primary; typeFilter=all
7	La Rambla Barcelona	PASS	76.249	65.114	count=5; first=La Rambla; mode=primary; typeFilter=street
8	Damrak Amsterdam	PASS	46.689	29.755	count=5; first=Damrak; mode=primary; typeFilter=all
9	Reeperbahn Hamburg	PASS	27.805	27.777	count=5; first=Reeperbahn; mode=primary; typeFilter=all
10	Maximilianstrasse Muenchen	PASS	65.220	44.848	count=5; first=Maximilianstraße; mode=primary; typeFilter=all

1.6 Optional Task 2: Questions About Missing Street or House Number

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	houses missing street count exists	PASS			housesMissingStreet=14526295
2	houses missing number count exists	PASS			housesMissingNumber=3365472
3	missing street percentage computable	PASS			13.96%
4	missing number percentage computable	PASS			3.23%
5	houses with admin areas counted	PASS			housesWithAdminAreas=103938360
6	houses from nodes counted	PASS			housesFromNodes=49868010
7	houses from ways counted	PASS			housesFromWays=54060704
8	houses from relations counted	PASS			housesFromRelations=138345
9	admin link density computable	PASS			houseAdminLinksPerHouse=5.58
10	missing fields not fabricated	PASS			missing values remain counted instead of filled with fake data

2 Sheet 2

2.1 Mandatory Task 1: Fast Point In Polygon Test

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	Stuttgart center	PASS	15.406	13.747	found=True; type=house; label=Straußweg 31; distanceM=13.76; adminAreas=7
2	Berlin Unter den Linden	PASS	40.005	20.322	found=True; type=house; label=Unter den Linden 1; distanceM=0.39; adminAreas=5
3	Munich Marienplatz	PASS	18.522	17.366	found=True; type=house; label=Marienplatz; distanceM=20.11; adminAreas=9
4	Hamburg Rathausmarkt	PASS	41.460	18.292	found=True; type=street; label=Rathausmarkt; distanceM=2.56; adminAreas=4
5	London Westminster	PASS	41.981	29.444	found=True; type=street; label=Bridge Street; distanceM=3.52; adminAreas=5
6	Paris Louvre	PASS	58.127	40.902	found=True; type=street; label=Ascenseur H; distanceM=11.46; adminAreas=9
7	Vienna Stephansplatz	PASS	37.424	13.263	found=True; type=house; label=Stephansplatz 3; distanceM=5.33; adminAreas=5
8	Zurich Bahnhofstrasse	PASS	53.850	55.072	found=True; type=street; label=Bahnhofstrasse; distanceM=3.88; adminAreas=7
9	Barcelona Eixample	PASS	30.044	19.093	found=True; type=house; label=Carrer del Consell de Cent 343; distanceM=29.24; adminAreas=9
10	Amsterdam Dam	PASS	23.474	22.731	found=True; type=street; label=Dam; distanceM=36.66; adminAreas=5

2.2 Mandatory Task 2: Save Administrative Attributes To Houses

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	Aalen Bahnhofstrasse 10	PASS	115.719	104.660	count=2; first=Bahnhofstraße 10; houseAdminAreas=6
2	Berlin Unter den Linden 1	PASS	23.060	22.860	count=2; first=Unter den Linden 1; houseAdminAreas=5
3	Muenchen Marienplatz 1	PASS	10.315	8.942	count=3; first=Marienplatz 1; houseAdminAreas=8
4	Hamburg Rathausmarkt 1	PASS	80.768	74.490	count=3; first=Rathausmarkt 1; houseAdminAreas=4
5	London Downing Street 10	PASS	173.885	162.174	count=1; first=Downing Street 10; houseAdminAreas=5
6	Paris Rue de Rivoli 99	PASS	13.688	12.525	count=1; first=Rue de Rivoli 99; houseAdminAreas=9
7	Wien Graben 1	PASS	4.436	2.501	count=3; first=Tiefer Graben 1-5; houseAdminAreas=4
8	Zuerich Bahnhofstrasse 1	PASS	97.266	83.347	count=3; first=Bahnhofstrasse 1; houseAdminAreas=9
9	Barcelona La Rambla 1	PASS	70.293	72.604	count=3; first=La Rambla 1; houseAdminAreas=9
10	Amsterdam Dam 1	PASS	117.417	93.338	count=3; first=Dam 1; houseAdminAreas=5

2.3 Mandatory Task 3: Reverse Geocoder

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	Stuttgart center	PASS	10.881	8.666	found=True; type=house; label=Straußweg 31; distanceM=13.76; adminAreas=7
2	Berlin Unter den Linden	PASS	30.208	10.654	found=True; type=house; label=Unter den Linden 1; distanceM=0.39; adminAreas=5
3	Munich Marienplatz	PASS	11.804	9.303	found=True; type=house; label=Marienplatz; distanceM=20.11; adminAreas=9
4	Hamburg Rathausmarkt	PASS	33.181	9.196	found=True; type=street; label=Rathausmarkt; distanceM=2.56; adminAreas=4
5	London Westminster	PASS	41.825	18.856	found=True; type=street; label=Bridge Street; distanceM=3.52; adminAreas=5
6	Paris Louvre	PASS	30.450	18.510	found=True; type=street; label=Ascenseur H; distanceM=11.46; adminAreas=9
7	Vienna Stephansplatz	PASS	13.326	10.789	found=True; type=house; label=Stephansplatz 3; distanceM=5.33; adminAreas=5
8	Zurich Bahnhofstrasse	PASS	14.857	12.065	found=True; type=street; label=Bahnhofstrasse; distanceM=3.88; adminAreas=7
9	Barcelona Eixample	PASS	14.277	11.879	found=True; type=house; label=Carrer del Consell de Cent 343; distanceM=29.24; adminAreas=9
10	Amsterdam Dam	PASS	16.679	14.125	found=True; type=street; label=Dam; distanceM=36.66; adminAreas=5

2.4 Mandatory Task 4: Frontend Reverse Geocoder Visualization

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	frontend click Stuttgart center	PASS	12.947	10.428	found=True; type=house; label=Straußweg 31; distanceM=13.76; adminAreas=7
2	frontend click Berlin Unter den Linden	PASS	13.970	10.964	found=True; type=house; label=Unter den Linden 1; distanceM=0.39; adminAreas=5
3	frontend click Munich Marienplatz	PASS	12.091	9.119	found=True; type=house; label=Marienplatz; distanceM=20.11; adminAreas=9
4	frontend click Hamburg Rathausmarkt	PASS	11.170	8.768	found=True; type=street; label=Rathausmarkt; distanceM=2.56; adminAreas=4
5	frontend click London Westminster	PASS	33.683	20.557	found=True; type=street; label=Bridge Street; distanceM=3.52; adminAreas=5
6	frontend click Paris Louvre	PASS	21.343	18.457	found=True; type=street; label=Ascenseur H; distanceM=11.46; adminAreas=9
7	frontend click Vienna Stephansplatz	PASS	12.912	10.328	found=True; type=house; label=Stephansplatz 3; distanceM=5.33; adminAreas=5
8	frontend click Zurich Bahnhofstrasse	PASS	13.739	11.164	found=True; type=street; label=Bahnhofstrasse; distanceM=3.88; adminAreas=7
9	frontend click Barcelona Eixample	PASS	33.189	10.379	found=True; type=house; label=Carrer del Consell de Cent 343; distanceM=29.24; adminAreas=9
10	frontend click Amsterdam Dam	PASS	35.307	14.490	found=True; type=street; label=Dam; distanceM=36.66; adminAreas=5

2.5 Optional Task 1: Attributes To All Other Objects

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	Koenigstrasse Stuttgart	PASS	44.068	41.811	count=5; first=Königstraße; type=street; nestedAdminAreas=11
2	Unter den Linden Berlin	PASS	80.677	78.507	count=5; first=Unter den Linden; type=street; nestedAdminAreas=5
3	Burger King Stuttgart	PASS	62.565	60.501	count=5; first=Burger King; type=poi; nestedAdminAreas=5
4	Lidl Muenchen	PASS	94.440	78.819	count=5; first=Lidl; type=poi; nestedAdminAreas=5
5	Eiffelturm Paris	PASS	13.462	10.825	count=5; first=Eiffelturm; type=poi; nestedAdminAreas=9
6	Tower Bridge London	PASS	43.122	40.489	count=5; first=Leonardo Royal Hotel London Tower Bridge; type=poi; nestedAdminAreas=4
7	Stephansdom Wien	PASS	24.319	3.620	count=5; first=Stephansdom Krypta; type=poi; nestedAdminAreas=5
8	Bahnhofstrasse Zuerich	PASS	365.155	353.058	count=5; first=Bahnhofstrasse; type=street; nestedAdminAreas=9
9	Sagrada Familia Barcelona	PASS	104.088	80.548	count=5; first=Leonardo Boutique Hotel Barcelona Sagrada Familia; type=poi; nestedAdminAreas=9
10	Damrak Amsterdam	PASS	31.683	31.642	count=5; first=Damrak; type=street; nestedAdminAreas=5

2.6 Optional Task 2: Objects In Current View Revisited

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	view Stuttgart /api/houses	PASS	7.982		houses=80; truncated=True
2	view Berlin /api/streets	PASS	21.947		streets=80; truncated=True
3	view Muenchen /api/admin	PASS	11.007		areas=80; truncated=True
4	view Hamburg /api/houses	PASS	6.716		houses=80; truncated=True
5	view Paris /api/streets	PASS	27.004		streets=80; truncated=True
6	view London /api/admin	PASS	15.971		areas=17; truncated=False
7	view Wien /api/houses	PASS	6.163		houses=80; truncated=True
8	view Zuerich /api/streets	PASS	22.382		streets=80; truncated=True
9	view Barcelona /api/admin	PASS	24.287		areas=72; truncated=False
10	view Amsterdam /api/houses	PASS	38.285		houses=80; truncated=True

2.7 Optional Task 3: Reverse Geocoder For Other Objects

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	other object Stuttgart center	PASS	22.482	9.488	found=True; type=house; label=Straußweg 31; distanceM=13.76; adminAreas=7
2	other object Berlin Unter den Linden	PASS	33.368	11.267	found=True; type=house; label=Unter den Linden 1; distanceM=0.39; adminAreas=5
3	other object Munich Marienplatz	PASS	29.837	9.551	found=True; type=house; label=Marienplatz; distanceM=20.11; adminAreas=9

4	other object Hamburg Rathausmarkt	PASS	10.523	8.850	found=True; type=street; label=Rathausmarkt; distanceM=2.56; adminAreas=4
5	other object London Westminster	PASS	29.740	18.967	found=True; type=street; label=Bridge Street; distanceM=3.52; adminAreas=5
6	other object Paris Louvre	PASS	21.790	20.108	found=True; type=street; label=Ascenseur H; distanceM=11.46; adminAreas=9
7	other object Vienna Stephansplatz	PASS	31.955	10.316	found=True; type=house; label=Stephansplatz 3; distanceM=5.33; adminAreas=5
8	other object Zurich Bahnhofstrasse	PASS	11.946	10.272	found=True; type=street; label=Bahnhofstrasse; distanceM=3.88; adminAreas=7
9	other object Barcelona Eixample	PASS	35.296	10.730	found=True; type=house; label=Carrer del Consell de Cent 343; distanceM=29.24; adminAreas=9
10	other object Amsterdam Dam	PASS	33.956	13.994	found=True; type=street; label=Dam; distanceM=36.66; adminAreas=5

2.8 Optional Task 4: Further Questions: Polygon Hierarchy and Intersections

Implementation status: Not implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	construct polygon hierarchy beyond stored parent links	NOT IMPLEMENTED			admin parent links exist, but constructive polygon hierarchy/overlay is not implemented
2	calculate polygon intersection points	NOT IMPLEMENTED			not implemented in current code path
3	construct new intersecting regions	NOT IMPLEMENTED			not implemented in current code path
4	construct new non-intersecting regions	NOT IMPLEMENTED			not implemented in current code path
5	export derived overlay polygons	NOT IMPLEMENTED			not implemented in current code path
6	intersection-aware PIP speedup proof	NOT IMPLEMENTED			not implemented in current code path
7	resolve conflicting overlapping admin polygons	NOT IMPLEMENTED			not implemented in current code path
8	return synthetic split polygons	NOT IMPLEMENTED			not implemented in current code path
9	test arbitrary polygon-polygon intersections	NOT IMPLEMENTED			not implemented in current code path
10	visualize constructed overlay hierarchy	NOT IMPLEMENTED			not implemented in current code path

3 Sheet 3

3.1 Mandatory Task 1: String Preprocessing

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	Koenigstrasse Stuttgart	PASS	54.414	39.932	count=5; first=Königstraße; mode=primary; typeFilter=all
2	Königstraße Stuttgart	PASS	63.250	41.801	count=5; first=Königstraße; mode=primary; typeFilter=all
3	Konigstrasse Stuttgart	WARN	83.728	65.240	count=2; first=Rue de Stuttgart; mode=primary; typeFilter=all; expected not found=Koenigstrasse
4	Konrad Adenauer Str Stuttgart	PASS	682.383	736.955	count=5; first=Konrad-Adenauer-Straße; mode=primary; typeFilter=all
5	Konrad-Adenauer-Strasse Stuttgart	PASS	656.982	709.607	count=5; first=Konrad-Adenauer-Straße; mode=primary; typeFilter=all
6	mUeNcHeN	PASS	43.526	33.729	count=5; first=München; mode=primary; typeFilter=all
7	München	PASS	33.211	32.921	count=5; first=München; mode=primary; typeFilter=all
8	Zuerich	PASS	52.639	54.174	count=5; first=Zürich; mode=primary; typeFilter=all
9	Zürich	PASS	45.443	46.398	count=5; first=Zürich; mode=primary; typeFilter=all
10	Gross Schauen	PASS	37.109	14.322	count=5; first=Groß Schauen; mode=primary; typeFilter=all

3.2 Mandatory Task 2: Geocoder

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	Stuttgart	PASS	40.092	41.035	count=5; first=Stuttgart; mode=primary; typeFilter=all
2	Berlin	PASS	62.785	65.566	count=5; first=Berlin; mode=primary; typeFilter=all
3	Muenchen	PASS	54.539	31.652	count=5; first=München; mode=primary; typeFilter=all
4	Hamburg	PASS	50.575	52.011	count=5; first=Hamburg; mode=primary; typeFilter=all
5	Frankfurt am Main	PASS	58.068	60.421	count=5; first=Frankfurt am Main; mode=primary; typeFilter=all
6	London	PASS	88.341	75.360	count=5; first=City of London; mode=primary; typeFilter=all
7	Paris	PASS	125.286	132.105	count=5; first=Paris; mode=primary; typeFilter=all
8	Barcelona	PASS	59.112	61.475	count=5; first=Barcelona; mode=primary; typeFilter=all
9	Amsterdam	PASS	51.340	43.922	count=5; first=Amsterdam; mode=primary; typeFilter=all
10	Zuerich	PASS	47.889	48.621	count=5; first=Zürich; mode=primary; typeFilter=all

3.3 Mandatory Task 3: Heuristics and Ordering

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	stuttgart	PASS	45.349	46.674	count=5; first=Stuttgart; mode=primary; typeFilter=all
2	Berlin Unter den Linden 1	PASS	20.625	20.075	count=2; first=Unter den Linden 1; mode=context; typeFilter=all
3	Muenchen Marienplatz 1	PASS	20.818	6.390	count=5; first=Marienplatz 1; mode=context; typeFilter=all
4	Koenigstrasse Stuttgart	PASS	40.072	41.016	count=5; first=Königstraße; mode=primary; typeFilter=all
5	Burger King Stuttgart	PASS	57.172	59.693	count=5; first=Burger King; mode=primary; typeFilter=all
6	Lidl Muenchen	PASS	91.582	73.623	count=5; first=Lidl; mode=primary; typeFilter=all
7	Eiffelturm Paris	PASS	36.073	10.678	count=5; first=Eiffelturm; mode=primary; typeFilter=all
8	Tower Bridge London	PASS	47.532	28.511	count=5; first=Leonardo Royal Hotel London Tower Bridge; mode=primary; typeFilter=all
9	Sagrada Familia Barcelona	PASS	84.403	67.866	count=5; first=Leonardo Boutique Hotel Barcelona Sagrada Familia; mode=primary; typeFilter=all
10	Damrak Amsterdam	PASS	40.052	26.096	count=5; first=Damrak; mode=primary; typeFilter=all

3.4 Mandatory Task 4: Timings, Measurements, Binary Save/Load

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
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1	binary snapshot loaded with embedded forward index	PASS			serverGeocodeIndexBuildMs=0.0
2	server spatial index build measured	PASS			serverSpatialIndexBuildMs=32692.074
3	forward context index terms measured	PASS			geocodeIndexTerms=3061667
4	forward primary index terms measured	PASS			geocodePrimaryIndexTerms=3049324
5	posting count measured	PASS			geocodeIndexPostings=1029909028
6	largest posting list measured	PASS			largestPostingList=23598165
7	house grid cells measured	PASS			houseGridCells=758520
8	street grid cells measured	PASS			streetGridCells=434038
9	admin grid cells measured	PASS			adminGridCells=30364
10	POI grid cells measured	PASS			poiGridCells=635732

3.5 Optional Task 1: Substring Search

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	berli	PASS	50.723	52.404	count=5; first=Berlin; mode=primary; typeFilter=all
2	muench	PASS	31.844	31.565	count=5; first=München; mode=primary; typeFilter=all
3	hambur	PASS	50.167	51.511	count=5; first=Hamburg; mode=primary; typeFilter=all
4	barcel	PASS	51.847	53.485	count=5; first=Barcelona; mode=primary; typeFilter=all
5	amster	PASS	57.577	34.299	count=5; first=Amsterdam; mode=primary; typeFilter=all
6	londo	PASS	112.401	117.656	count=5; first=Groß-London; mode=primary; typeFilter=all
7	zueri	PASS	15.421	14.455	count=5; first=Zürich; mode=primary; typeFilter=all
8	frankf	PASS	118.341	110.217	count=5; first=Frankfurt am Main; mode=primary; typeFilter=all
9	stuttg	PASS	26.689	26.203	count=5; first=Stuttgart; mode=primary; typeFilter=all
10	koenigstr	PASS	57.690	22.330	count=5; first=Königstraße; mode=primary; typeFilter=all

3.6 Optional Task 2: Aggregating Nearby Results

Implementation status: Not implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	cluster nearby result example 1	NOT IMPLEMENTED			not implemented in current code path
2	cluster nearby result example 2	NOT IMPLEMENTED			not implemented in current code path
3	cluster nearby result example 3	NOT IMPLEMENTED			not implemented in current code path
4	cluster nearby result example 4	NOT IMPLEMENTED			not implemented in current code path
5	cluster nearby result example 5	NOT IMPLEMENTED			not implemented in current code path
6	cluster nearby result example 6	NOT IMPLEMENTED			not implemented in current code path
7	cluster nearby result example 7	NOT IMPLEMENTED			not implemented in current code path
8	cluster nearby result example 8	NOT IMPLEMENTED			not implemented in current code path
9	cluster nearby result example 9	NOT IMPLEMENTED			not implemented in current code path
10	cluster nearby result example 10	NOT IMPLEMENTED			not implemented in current code path

3.7 Optional Task 3: Error Correction or Fuzzy Search

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	stuttgart	PASS	48.390	37.014	count=5; first=Stuttgart; mode=primary; typeFilter=all
2	stuttgrat	PASS	26.355	25.581	count=5; first=Stuttgart; mode=primary; typeFilter=all
3	berln	PASS	38.771	24.299	count=5; first=Berlin; mode=primary; typeFilter=all
4	munchn	PASS	31.091	31.018	count=5; first=München; mode=primary; typeFilter=all
5	hambug	PASS	19.266	18.544	count=5; first=Hamburg; mode=primary; typeFilter=all

6	frankfrut	PASS	33.315	33.776	count=5; first=Frankfurt am Main; mode=primary; typeFilter=all
7	londn	PASS	85.880	89.403	count=5; first=Groß-London; mode=primary; typeFilter=all
8	pariis	PASS	117.467	124.061	count=5; first=Paris; mode=primary; typeFilter=all
9	barcelona	PASS	54.131	56.024	count=5; first=Barcelona; mode=primary; typeFilter=all
10	amsterdm	PASS	30.551	30.566	count=5; first=Amsterdam; mode=primary; typeFilter=all

3.8 Optional Task 4: Results Only In Current View

Implementation status: Not implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	strict viewport-only geocode for Stuttgart while map is in Barcelona	NOT IMPLEMENTED			geocoder accepts viewport/focus for ranking, but does not enforce results-only-in-view
2	strict viewport-only geocode for Berlin while map is in London	NOT IMPLEMENTED			geocoder can return outside-view results
3	strict viewport-only brand search in current view	NOT IMPLEMENTED			not implemented in current code path
4	empty-view fallback rule when no current-view result exists	NOT IMPLEMENTED			not implemented in current code path
5	frontend switch for only-current-view filtering	NOT IMPLEMENTED			not implemented in current code path
6	API flag that rejects outside-view candidates	NOT IMPLEMENTED			not implemented in current code path
7	viewport-only deterministic street search	NOT IMPLEMENTED			not implemented in current code path
8	viewport-only POI search	NOT IMPLEMENTED			not implemented in current code path
9	viewport-only address search	NOT IMPLEMENTED			not implemented in current code path
10	viewport-only natural-language search	NOT IMPLEMENTED			not implemented in current code path

3.9 Optional Task 5: Example Natural Queries

Implementation status: Implemented

No.	Check / query	Verdict	Wall ms	Server ms	Observed result
1	Aalen Bahnhofstrasse 10	PASS	6649.323	99.304	count=2; first=Bahnhofstraße 10; intent=address; usedLlm=True; verified=True; passes=2
2	Stuttgart Burger King	PASS	2811.881	85.245	count=5; first=Burger King; intent=named_poi_in_place; usedLlm=True; verified=True; passes=2
3	Closest Park to Kaistrasse 5 Kiel	PASS	2034.669	2202.410	count=5; first=THW Kiel; intent=nearest_category_to_address; usedLlm=True; verified=True; passes=2
4	where can i find burger king in stuttgart	PASS	3237.514	77.833	count=5; first=Burger King; intent=named_poi_in_place; usedLlm=True; verified=True; passes=2
5	wo finde ich lidl in muenchen	PASS	2693.196	81.931	count=5; first=Lidl; intent=named_poi_in_place; usedLlm=True; verified=True; passes=2
6	where can i buy shampoo near Marienplatz Munich	PASS	1644.428	1762.242	count=5; first=Drogerie Wittelsbach; intent=nearest_concept_to_address; usedLlm=True; verified=True; passes=2
7	wo bekomme ich brot am Alexanderplatz Berlin	PASS	1532.398	1532.510	count=5; first=Zeit für Brot; intent=nearest_concept_to_address; usedLlm=True; verified=True; passes=2
8	where can I find a pharmacy near Eiffel Tower Paris	PASS	3209.443	1805.634	count=5; first=Pharmacy; intent=named_poi_in_place; usedLlm=True; verified=True; passes=2
9	wo finde ich eine tankstelle in Hamburg	PASS	1256.479	1341.587	count=5; first=Freie Tankstelle; intent=nearest_category_to_address; usedLlm=True; verified=True; passes=2
10	where can i find a coffee shop in Amsterdam	PASS	1736.773	542.089	count=5; first=Coffee Shop Amsterdam; intent=named_poi_in_place; usedLlm=True; verified=True; passes=2

Fuzzy Search Diagnosis

The current fuzzy search path does not use the old vocabulary-size guard. It creates seed substrings, finds candidate vocabulary terms through the suffix array, verifies them with bounded edit distance, and then lets the normal geocoder

ranking score the verified terms. This keeps Europe-scale typo correction bounded without a full vocabulary scan per query.

Audit Notes

Rows marked PASS satisfied the concrete runtime or statistics check. WARN means the endpoint returned usable data, but the ranking or normalization did not satisfy the stricter expectation in that row. FAIL means the expected behavior was not observed. NOT IMPLEMENTED means the final program does not include that optional task as a feature.